DESIGN CHALLENGE

Project Specifications

Teams

Form groups of 4 ± 1 students. All members of the group are equally responsible for the project. The instructor reserves the right to award different grades for different group members in situations of obviously disproportionate participation.

Performance Specifications

The objective of this project is to build a small-scale elevator. The elevator should contain 4 floors. Each floor must be equipped with a call button (button outside the elevator). The elevator car should also have a 4-button control panel for sending the elevator to the desired floors and a door open button which will keep the elevator door open while held (this simulates the panel inside the elevator car). Passengers should be capable of calling the elevator to a specific floor and specifying a floor to which to travel. Upon reaching the floor, an actuator (pneumatic or solenoid) should simulate the opening and closing of the door. The "door" must remain open for three seconds to permit passengers to enter and exit the car and should remain open if someone is in the way. For bonus points, a buzzer should sound, and a light should flash if the door is held open beyond the three seconds for greater than 15 seconds. Further, the elevator should function as a real elevator – should stop at activated floors as they pass rather than traveling to floors in the order in which they are pressed. When not in use, the elevator should return to the home position – floor 1.

Elevator should be capable of lifting and delivering a payload of $6\ oz$ between floors. Elevator must be capable of traveling fully unloaded as well as fully loaded, and quantities in between.

The opening of each floor should be at least 4 inches high by 3 inches wide.

A vertical height of at least 1 inch should separate each floor. Height above top floor and below bottom floor is arbitrary if entire project meets total size limitations.

Elevator should be capable of lifting and delivering a payload of $6\ oz$ between floors. Elevator must be capable of traveling fully unloaded as well as fully and partially loaded.

Physical Dimensions

- Maximum height: 36 inches
- Maximum base size: 24 in x 24 in
- Should fit on desktop or at lab station comfortably for demonstration. One student should be able to lift the elevator (with laptop not connected). If two students are required, it is too large!
- Laptop, controllers, etc. are not included in these dimensions IF they can be easily removed from the platform for moving the project.

Power Restrictions

Power supplies are available in the lab to support 24V, 12V, and 5V DC, 5 A or less or standard 110V AC. Any other power supply must be supplied by the group and approved by instructor. All power supplies must be used and configured safely to avoid risk of shock. Batteries are not recommended as this is not a mobile project, but they are not prohibited if they are stored, charged, and used in a safe and responsible manner.

Limitations

Projects may not consist of more than 50% kit parts (by volume and by quantity). All projects must make use of at least one feature that does not come from a kit product. Kit parts are defined as VEX, Mindstorms/Lego, Tetrix, K'NEX, etc.

Materials such as buttons, LEDs, Arduinos, breadboards, etc. may be borrowed. Reasonable requests for items to be purchased from Amazon will be considered. Keep in mind, however, that our shipping is very slow, despite Amazon two-day shipping. Materials will be signed out to the members of a group. Loaned parts and any materials purchased with MTR funds are expected to be returned at the conclusion of the project in working order in the condition lent, less reasonable wear-and-tear.

Students may use any (safe, nonhazardous) materials to construct their system, including art supplies. While neat work is expected, this project is not graded on construction quality but rather on functionality and engineering design.

Students may, with supervision, use the equipment in either the clean prototyping (COO 111) or light fabrication (COO 145) laboratories.

Instructor has final decision for any questionable materials, electronics, etc. used in the project.

Documentation Requirements and Rubric

Documentation (150pt)

Report must adhere to the report format and guidelines. Additionally, the formal report must include:

- Full mechanical drawings and/or assembly using a professional CAD program (Autodesk Inventor or Fusion 360 is recommended as it permits easiest collaboration)
- Bill of Materials and budget/cost, including cost of borrowed materials
- Electrical schematic (MultiSim or Fritzing recommended, must be electronic)
- Block diagram of control system
- Flowchart of programming logic
- Data sheets must be included for all electronics used in the project, regardless of source

Oral Presentation (50pt)

Groups are expected to present their design to the class at the conclusion of the semester, utilizing PowerPoint and any other visual aids necessary. All group members must participate in the presentation to receive credit.

Project Demonstration (100pt)

In addition to an in-lab demonstration at the conclusion of the semester, groups must also submit a short video highlighting and illustrating the functionality of their project. Phone camera recordings are acceptable and encouraged. Please use a general file format such as MP4 to prevent compatibility issues when viewing.

Maximum *total* grade for a non-functional project is limited to a "C" -- requiring excellent submission materials and well-documented engineering design in spite of overall lack of functionality of the final product.

Bonus Opportunities

A 5% bonus will be awarded to the team whose elevator correctly completes a sequence of floor changes (announced at demonstration) and can stop within 1 mm of the floor consistently in the shortest amount of time. Additional bonus will be considered by the instructor for projects exhibiting exceptional and unique design, above-and-beyond features, and excellent performance.